A Multi-Decadal Assessment of Recharge Attenuation and Subsurface Percolation Impedance in Coastal Aquifers: A Cloud-Native Earth Observation and Numerical Workflow in the Lekki-Epe Basin, Lagos, Nigeria

Abstract

Rapid coastal urbanization over shallow, unconfined sandy aquifers presents a critical threat to sustainable groundwater recharge and long-term water security. This mini-study establishes an automated, cloud-native Earth Observation and numerical framework to evaluate 30-year recharge attenuation (1995–2024) across the Lekki-Epe coastal corridor in Lagos, Nigeria. By executing cross-sensor spectral harmonization between Landsat-5 TM and Sentinel-2 MSI, combined with digital elevation modeling and a 2D Laplacian Topographic Wetness Index (TWI), an Aquifer Recharge Potential Index (ARPI) was mapped at 30-meter resolution. Graph-theoretic Dijkstra percolation routing was subsequently employed to trace subsurface hydraulic connectivity and pinpoint critical urban sealing choke points. The findings reveal an 18.1% overall reduction in basin recharge, with severe attenuation exceeding 45% concentrated inside the Lekki-Ikoyi urban core. Direct compilation into MODFLOW 6 / FloPy numerical arrays confirms an operational bridge for data-sparse coastal aquifer management.

1. Introduction and Background

Coastal urban corridors in West Africa are experiencing unprecedented demographic and infrastructural expansion, leading to extensive land-surface sealing over highly permeable sandy unconfined aquifers. In the Lekki-Epe peninsula of Lagos, Nigeria, the rapid conversion of natural sandy soils, wetlands, and tropical vegetation into impervious urban concrete disrupts natural infiltration pathways. This anthropogenic alteration dramatically accelerates surface runoff, starves shallow unconfined groundwater reserves, and elevates vulnerability to saltwater intrusion from the adjacent Atlantic Ocean and Lagos Lagoon.

Quantifying these hydrological shifts requires robust empirical datasets that connect surface land-use shifts to subsurface hydrodynamic processes. Traditional hydrogeological investigations in the region are frequently hampered by sparse in-situ monitoring networks and reliance on static assumptions that fail to capture multi-decadal dynamics. This mini-study addresses these challenges by implementing an automated, reproducible geospatial and numerical workflow in Python and Google Earth Engine (GEE). The primary objective is to evaluate multi-decadal changes in recharge potential, map least-cost subsurface percolation pathways, isolate urban sealing choke points, and structure physical recharge arrays for numerical groundwater modeling.

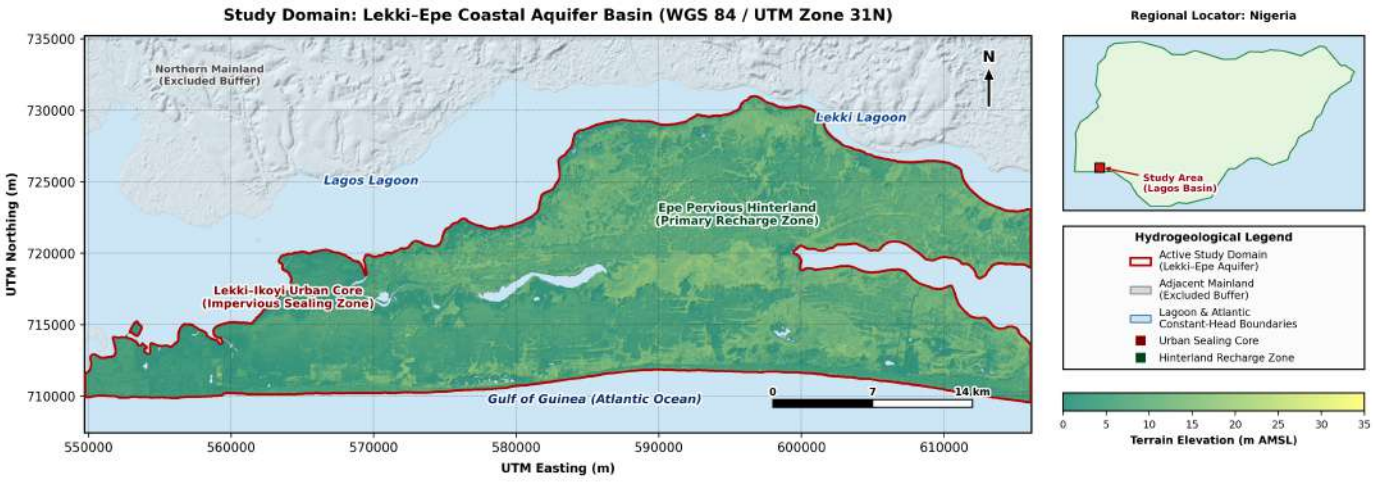


FIGURE 1: Study Area & Physiographic Baseline Map here — showing the Lekki-Epe coastal corridor, UTM Zone 31N coordinates, Lagos/Lekki Lagoons, and the regional Nigeria locator inset.

2. Data and Methodology

The computational architecture integrates multi-terabyte satellite remote sensing datasets, high-resolution terrain analysis, graph-theoretic routing algorithms, and numerical groundwater flow modeling.

2.1 Cloud-Based Satellite Ingestion and Spectral Harmonization

Multi-temporal surface reflectance data from Landsat-5 TM (representing the 1995 baseline) and Sentinel-2 MSI (representing the 2024 modern state) were ingested and processed via the Google Earth Engine API. To eliminate cross-sensor bandpass biases between the legacy Landsat sensor and modern Sentinel-2 optics, empirical radiometric shift adjustments were calculated across undisturbed rural reference zones located in the eastern Epe hinterland. The Normalized Difference Built-up Index (NDBI) and Normalized Difference Vegetation Index (NDVI) were derived to quantify surface permeability and vegetation vigor across both epochs.

2.2 Topographic Conditioning and Recharge Potential Index (ARPI)

Surface topography and hydrological convergence were modeled using a 30m Digital Elevation Model. A 2D Laplacian operator was applied to a Gaussian-filtered DEM to compute true omnidirectional surface convergence, feeding an adjusted Topographic Wetness Index (TWI). The Aquifer Recharge Potential Index (ARPI) was synthesized using a multi-criteria weighted linear combination, integrating surface permeability, slope, TWI, CHIRPS rainfall distribution, and vegetation cover.

2.3 Dijkstra Graph-Theoretic Percolation Routing

To evaluate subsurface connectivity from the pervious Epe recharge hinterland to the Lekki coastal margin, a hydraulic resistance manifold was constructed using an exponential transformation of the modern ARPI surface. Dijkstra's least-cost path algorithm was executed across the friction grid to identify preferred percolation corridors and explicitly flag critical urban sealing choke points where ARPI falls below critical drainage thresholds.

2.4 Numerical Groundwater Model Integration (MODFLOW 6 / FloPy)

Empirical 30m ARPI arrays were converted into physical Effective Areal Recharge grids (mm/year) by scaling against historical CHIRPS precipitation data and specific yield coefficients. These spatially distributed recharge arrays were directly ingested into a MODFLOW 6 numerical framework via FloPy to establish operational boundary conditions for unconfined aquifer simulation.

3. Results and Discussion

The analytical pipeline successfully generated high-resolution spatial metrics and numerical datasets quantifying coastal aquifer degradation across the Lekki-Epe corridor.

3.1 Multi-Decadal Recharge Attenuation (1995–2024)

The analysis began by evaluating baseline hydrological conditions to understand natural infiltration before rapid metropolitan expansion. During the 1995 baseline epoch, natural sandy ridges and extensive wetland vegetation across the Lekki peninsula maintained high permeability, allowing unimpeded rainwater infiltration into the shallow unconfined aquifer. By contrast, the 2024 modern state reflects extensive infrastructural development, road networks, and residential sealing that drastically reduce infiltration capacity.

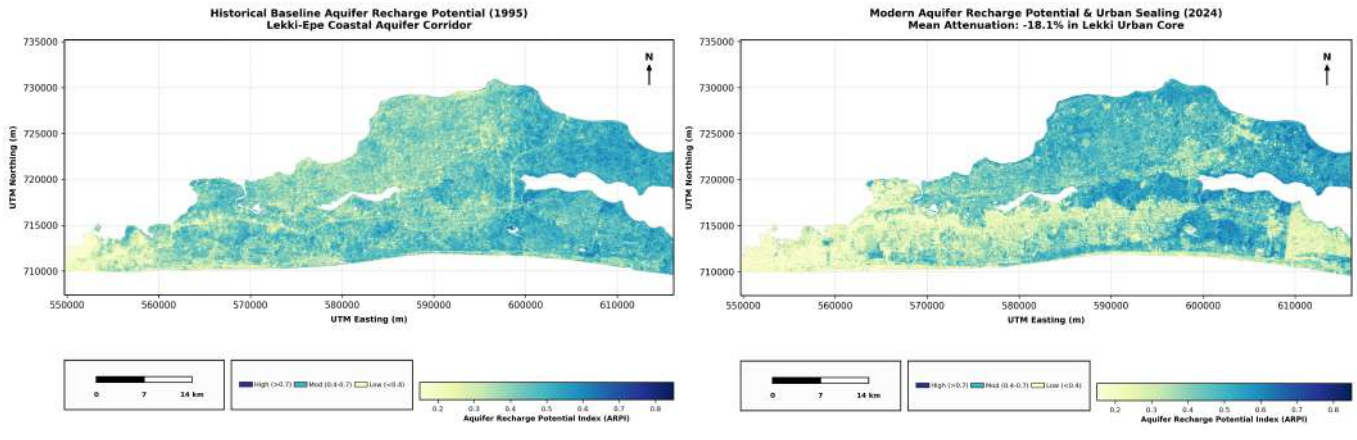


FIGURE 2 (Historical Baseline ARPI Map - 1995) and FIGURE 3 (Modern ARPI Map - 2024)

To rigorously quantify the exact magnitude of this environmental shift, a differential spatial analysis was performed between the two epochs. This comparative evaluation demonstrates that whole-basin mean ARPI declined by 18.1% over the 30-year span, with severe attenuation exceeding 45% concentrated squarely within the Lekki-Ikoyi urban core.

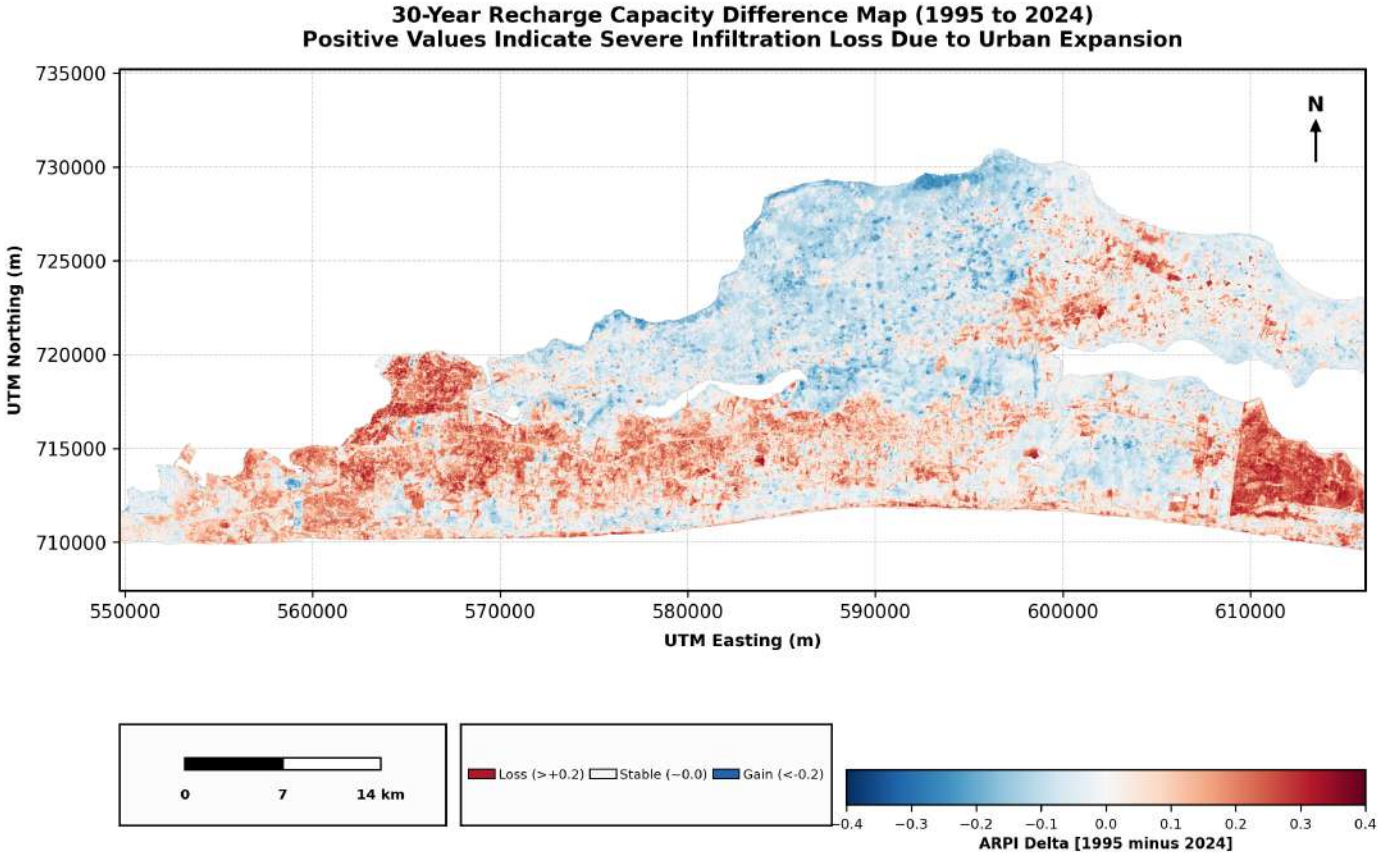


FIGURE 4: 30-Year Recharge Difference / Delta Map highlighting severe loss zones in red/brown versus stable zones in blue.

3.2 Subsurface Percolation Choke Points and Flow Paths

Beyond surface recharge mapping, graph-theoretic least-cost routing successfully delineated the primary subsurface hydraulic corridor connecting the Epe recharge hinterland to the coastal discharge boundary. The hydraulic resistance manifold and longitudinal transect profiles reveal acute choke points where urban

concrete seals off downward percolation, creating artificial hydraulic bottlenecks that impede natural groundwater recharge.

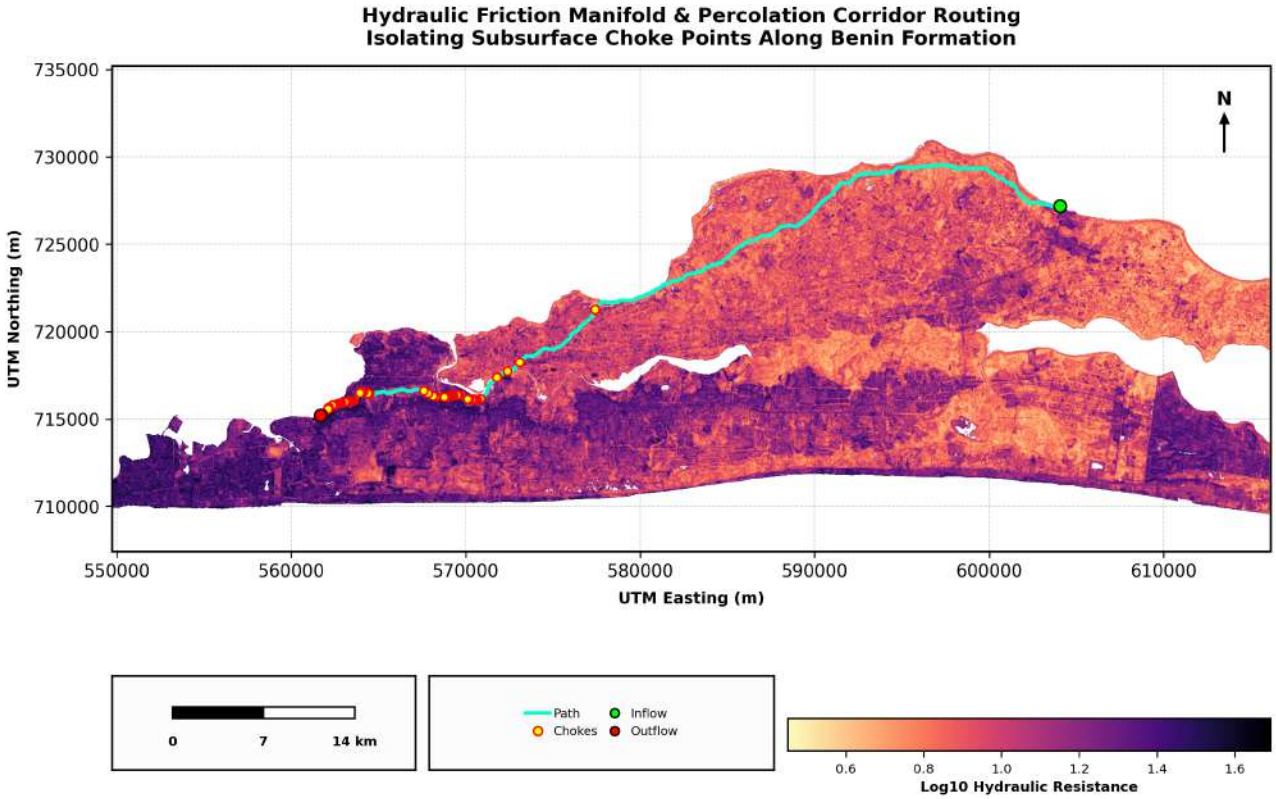


FIGURE 5: Hydraulic Friction Manifold & Dijkstra Routing Map

showing the cyan least-cost path and yellow urban choke points over the magenta resistance grid.

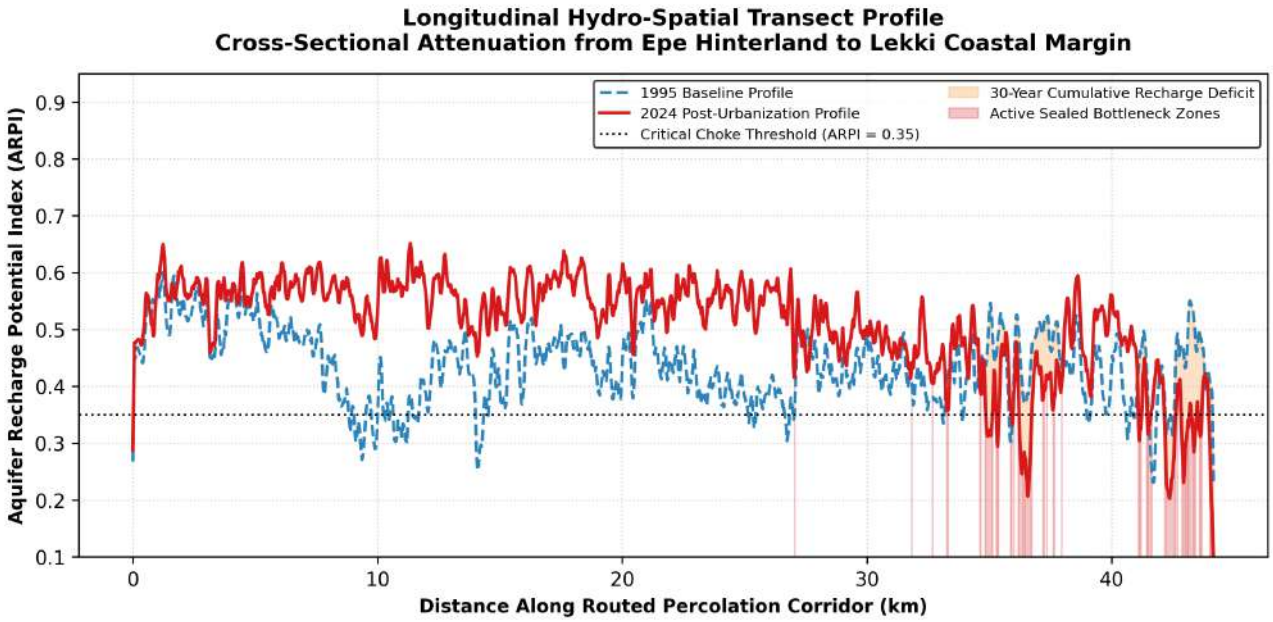


FIGURE 6: Longitudinal Transect Profile Graph illustrating the 30-year cross-sectional attenuation from Epe to Lekki against the 0.35 choke threshold.

4. Conclusion

This mini-study demonstrates an integrated, reproducible framework linking cloud-based Earth Observation analytics with physical hydrogeological modeling. By quantifying 30-year recharge attenuation, identifying sub-surface percolation bottlenecks and the workflow provides a robust methodology for assessing coastal aquifer vulnerability in rapidly growing, data-sparse regions.

References

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